

Recap of
9/11 class

$$Y = X\beta + \epsilon$$

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{(p-1)1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{(p-1)n} \end{bmatrix}_{n \times p}$$

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{p-1} \end{bmatrix}_p$$

where, p = # of predictors in our model

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} Q(\beta) = \underset{\beta}{\operatorname{argmin}} (Y - X\beta)^T (Y - X\beta)$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y \quad \text{assuming } X$$

$$y = \beta_0 + \beta_1 x + \epsilon$$

$$\epsilon = y - \beta_0 - \beta_1 x$$

$$Y = X\beta + \epsilon$$

$$Q = (Y - X\beta)^T (Y - X\beta)$$

LS

$$\epsilon = Y - X\beta$$

$$\frac{\partial Q}{\partial \beta} = 0$$

$$Q = \sum \epsilon_i^2$$

$$Q = \sum (y_i - \beta_0 - \beta_1 x_i)^2$$

$$\frac{\partial Q}{\partial \beta_0} = 0 \quad \frac{\partial Q}{\partial \beta_1} = 0$$

 $n=2$

$$Y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \end{bmatrix}$$

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

$$Q = \sum_{i=1}^{n=2} (y_i - \beta_0 - \beta_1 x_i)^2$$

$$= (y_1 - \beta_0 - \beta_1 x_1)^2 + (y_2 - \beta_0 - \beta_1 x_2)^2$$

$$Q = (Y - X\beta)^T (Y - X\beta)$$

$$A^T A = \sum_{i=1}^n a_i^2$$

$$\begin{bmatrix} y_1 - \beta_0 - \beta_1 x_1 \\ y_2 - \beta_0 - \beta_1 x_2 \end{bmatrix}^T \begin{bmatrix} y_1 - \beta_0 - \beta_1 x_1 \\ y_2 - \beta_0 - \beta_1 x_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 - \beta_0 - \beta_1 x_1 & y_2 - \beta_0 - \beta_1 x_2 \end{bmatrix} \begin{bmatrix} y_1 - \beta_0 - \beta_1 x_1 \\ y_2 - \beta_0 - \beta_1 x_2 \end{bmatrix}$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

 $\hat{Y} \rightarrow$ prediction

$$\hat{Y} = X\hat{\beta} = X(X^T X)^{-1} X^T Y$$

$$\hat{Y} = HY$$

claim: H is a projection matrixfor any matrix M to be projection matrix it has to be

$$1) \text{ symmetric } M^T = M$$

$$2) \text{ idempotent } M^2 = M$$

$$1) H^T = H$$

$$\begin{aligned} H^T &= (X(X^T X)^{-1} X^T)^T \\ &= X((X^T X)^{-1})^T X^T \\ &= X(X^T X)^{-1} X^T \\ &= X(X^T X)^{-1} X^T \\ &= H \end{aligned}$$

$$2) H^2 = H$$

$$\begin{aligned} H^2 &= (X(X^T X)^{-1} X^T)^2 \\ &= X(X^T X)^{-1} (X^T X)^{-1} X^T X^T \\ &= X(X^T X)^{-1} X^T \\ &= H \end{aligned}$$

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

$$= \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} \begin{bmatrix} 1 \\ x_i \end{bmatrix}$$

$$\frac{\partial (\hat{\beta}^T A)}{\partial \beta} = A$$

$$\frac{\partial C}{\partial \beta} = 0$$

$$\frac{\partial (\hat{\beta}^T A \beta)}{\partial \beta} = 2A\beta$$

$$\frac{\partial Q}{\partial \beta} = \frac{\partial}{\partial \beta} (Y - X\beta)^T (Y - X\beta)$$

$$0 = \frac{\partial}{\partial \beta} (Y^T Y - Y^T X\beta - \beta^T X^T Y + \beta^T X^T X \beta)$$

$$0 = \frac{\partial}{\partial \beta} (-2Y^T X\beta + \beta^T X^T X \beta)$$

$$0 = -2X^T Y + 2X^T X \beta$$

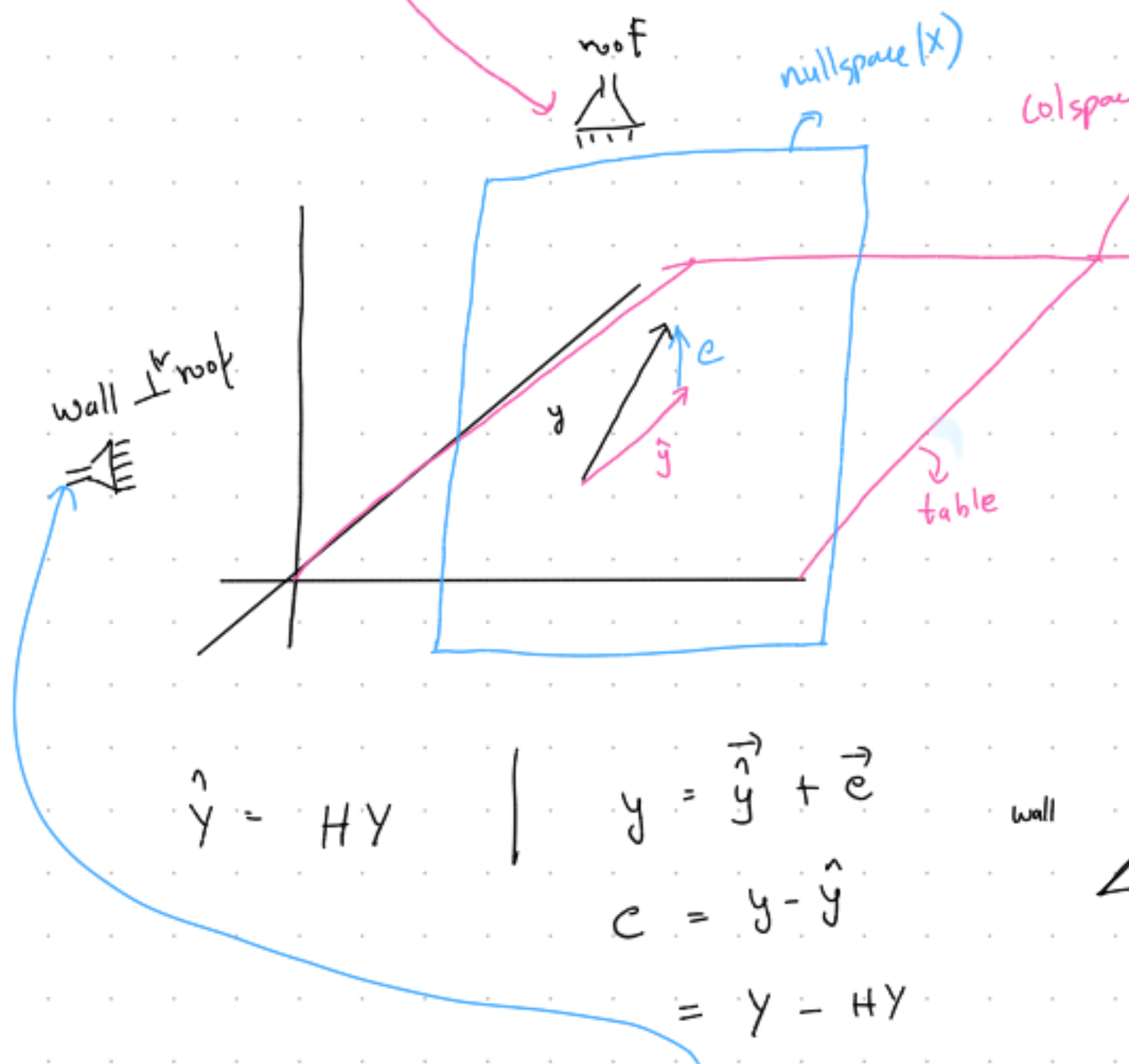
$$X^T X \beta = X^T Y$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

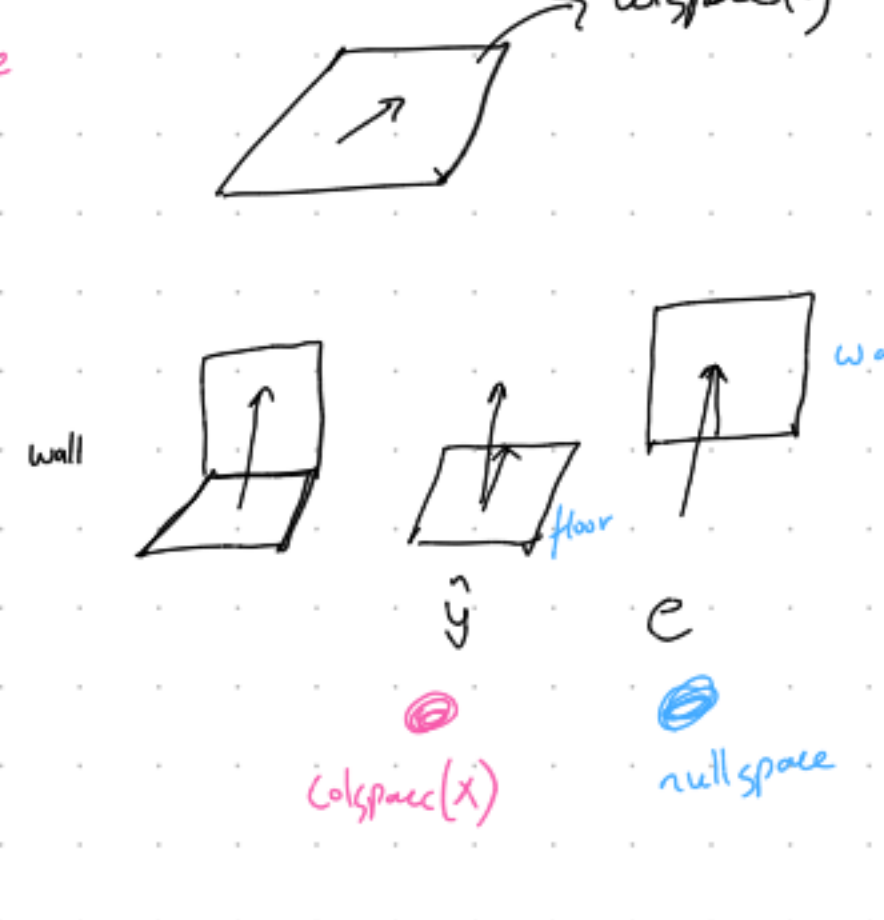
$$(ABC)^T = C^T B^T A^T$$

$$(A^T)^T = (A)^T$$

$$A^T A = I$$



$$\begin{aligned} \hat{Y} &= HY \\ Y &= \hat{Y} + e \\ e &= Y - \hat{Y} \\ e &= (I - H)Y \end{aligned}$$

 $(I - H)$ is projection !!

$$1) (I - H)^T = I^T - H^T = I - H$$

$$\begin{aligned} 2) (I - H)^2 &= (I - H)(I - H) = (II - IH - HI + HH) \\ &= I - H - H + H \\ &= (I - H) \end{aligned}$$

We want to estimate σ^2 how?

$$\begin{aligned} SSE &= \sum \epsilon_i^2 = e^T e \\ &= ((I - H)Y)^T (I - H)Y \\ &= Y^T (I - H)^T (I - H)Y \\ &= Y^T (I - H)Y \\ &= Y^T (I - H)Y \end{aligned}$$

$$Y \sim N(X\beta, \sigma^2 I)$$

$$e = (I - H)Y \sim N(0, \sigma^2 (I - H))$$

$$E[(I - H)Y] = E[X\beta - HX\beta] = E[X\beta - X(X^T X)^{-1} X^T X\beta] = E[X\beta - X\beta] = 0$$

$$\begin{aligned} \text{Var}[(I - H)Y] &= (I - H) \text{Var}(Y) (I - H)^T \\ &= (I - H) \sigma^2 I (I - H)^T \\ &= \sigma^2 (I - H) \end{aligned}$$

$$\text{Var}(Ax) = A \text{Var}(x) A^T$$

What is χ^2_{df} ? (hi - squared $\Rightarrow$ t-test)We say a R.V. V follows χ^2_{df} if

$$V = Z_1^2 + Z_2^2 + \dots + Z_n^2$$

where $Z_i \sim N(0, 1)$

$$e \sim N(0, \sigma^2 (I - H))$$

$$\frac{e}{\sigma} \sim N(0, \frac{\sigma^2 (I - H)}{\sigma^2})$$

divide by $(I - H)^{1/2}$

$$\frac{(I - H)^{-1/2} e}{\sigma} \sim N(0, I) \stackrel{\text{def}}{=} \chi^2_{df}$$

$$\frac{(I - H)^{-1/2} (I - H)Y}{\sigma} \sim N(0, I)$$

$$\frac{(I - H)^{-1/2} Y}{\sigma} \sim N(0, I)$$

$$\left\| \frac{(I - H)^{-1/2} Y}{\sigma} \right\|^2 \sim \chi^2_{df}$$

where $df = \text{Rank of } (I - H)$

$$\text{Rank of } (I - H) = \text{tr}(I - H) = \text{tr}(I) - \text{tr}(H)$$

$$= n - \text{tr}(X(X^T X)^{-1} X^T)$$

$$= n - \text{tr}(X^T X (X^T X)^{-1})$$

$$= n - \text{tr}(I_p)$$

$$= n - p$$

$$\text{Rank of } I - H = n - p$$

$$\left\| \frac{(I - H)^{-1/2} Y}{\sigma} \right\|^2 \sim \chi^2_{n-p}$$

$$\sum \epsilon_i^2$$

$$\frac{SSE}{\sigma^2} = \left\| \frac{e}{\sigma} \right\|^2 = \left\| \frac{(I - H)Y}{\sigma} \right\|^2 \sim \chi^2_{n-p}$$

$$E\left[\frac{SSE}{\sigma^2}\right] = n - p$$

$$E\left[\frac{SSE}{n-p}\right] = \sigma^2$$

$$E[\hat{\sigma}^2] = \sigma^2 \quad \text{unbiased}$$

$$\hat{\sigma}^2 = \frac{e^T e}{n-p}$$

What is norm?

$$v = v_1 i + v_2 j$$

$$v = [v_1 \ v_2]$$

What is norm

$$\|v\| = \sqrt{v_1^2 + v_2^2}$$

$$v = [v_1 \ v_2 \ v_3 \ v_4 \ \dots \ v_n]$$

$$\|v\| = \sqrt{v_1^2 + v_2^2 + v_3^2 + \dots + v_n^2}$$

we can multiply if the dimensions match

$$n \times 1 \cdot 1 \times p \Rightarrow n \times p$$

$$(I - H) \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \end{bmatrix}_{n \times n}$$

$$\text{tr}(I_n)$$

$$\text{tr}(ABC) = \text{tr}(BCA)$$

$$= \text{tr}(CAB)$$

$$X_{n \times p}$$

$$E(V) = df$$

$$\text{when } v \sim \chi^2_{df}$$